

§14.3 Partial Derivatives

Homework : 5,15,17,29,37,43,51,55,61,83,88,89,90,91

(1) Partial Derivatives (偏導數) $f_x(a,b)$ of f with respect to x at (a,b) .

Definition :

$$(i) f_x(a,b) = \lim_{h \rightarrow 0} \frac{f(a+h,b) - f(a,b)}{h}$$

(ii) 同理可定義對 y 的偏導數作用在 (a,b) 為

$$f_y(a,b) = \lim_{h \rightarrow 0} \frac{f(a,b+h) - f(a,b)}{h}.$$

Notation : $\frac{\partial f}{\partial x} = f_x = f_1$; $\frac{\partial f}{\partial y} = f_y = f_2$

Example 1 : $f(x,y) = x^2 + xy + y^2$

$$\Rightarrow f_x = 2x + y \quad (\text{把 } y \text{ 當常數對 } x \text{ 微分})$$

$$f_y = 2y + x \quad (\text{把 } x \text{ 當常數對 } y \text{ 微分})$$

$$f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = f_{11} = 2$$

$$(f_x)_y = f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = f_{12} = 1$$

$$(f_y)_x = f_{yx} = f_{21} = 1$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = f_{22} = 2$$

(2)

(i) Clairau's Theorem : If f_{xy} and f_{yx} are continuous on a region D , then

$$f_{xy}(a,b) = f_{yx}(a,b) \quad \text{for any } (a,b) \in D.$$

(ii) $f_{xyy} = \left((f_x)_y \right)_y$, f_{yxy} , f_{yyx} are continuous on D , then

$$f_{xyy}(a,b) = f_{yxy}(a,b) = f_{yyx}(a,b) \text{ for } (a,b) \in D.$$

Example 2 : Let $u(x,y) = e^x \sin y$. Prove that $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Proof :

$$u_x = e^x \sin y \quad u_y = e^x \cos y$$

$$u_{xx} = e^x \sin y \quad u_{yy} = -e^x \sin y$$

$$\Rightarrow u_{xx} + u_{yy} = 0 \text{ (Laplace's Equation)}$$

Example 3 : Let $u(x,t) = f(x-at)$. Prove that $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$.

Proof :

$$u_t = -af'(x-at) \quad u_{tt} = a^2 f''(x-at)$$

$$u_x = f'(x-at) \quad u_{xx} = f''(x-at)$$

$$\Rightarrow u_{tt} = a^2 u_{xx} \text{ (Wave's Equation)}$$

Example 4 : Let $xyz = \cos(x+y+z)$. Find $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$.

Solution :

將 z 想成以 x 和 y 為獨立變數的函數

$$yz + xy \frac{\partial z}{\partial x} = (-\sin(x+y+z)) \left(\frac{\partial z}{\partial x} + 1 \right)$$

$$\Rightarrow \frac{\partial z}{\partial x} = \frac{-\sin(x+y+z) - yz}{xy + \sin(x+y+z)}$$

$$\xRightarrow{\text{同理}} \frac{\partial z}{\partial y} = \frac{-\sin(x+y+z) - xz}{xy + \sin(x+y+z)}$$

Example 5 : $f(x, y) = (x^3 + y^3)^{\frac{1}{3}} e^{-\sin y}$. Find $f_x(1, 0)$.

Solution :

$$f(x, 0) = \sqrt[3]{x^3} = x \Rightarrow f'(x) = 1$$

$$\Rightarrow f_x(1, 0) = 1. \text{ (} f_x(1, 0) \text{ 將 } y \text{ 固定為 } 0 \text{ 對 } x \text{ 的微分.)}$$

Example 6 :

$$f(x, y) = \begin{cases} \frac{x^3 y - xy^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

(1) Find $f_x(x, y)$, $f_y(x, y)$ for $(x, y) \neq (0, 0)$.

(2) Find $f_x(0, 0)$ and $f_y(0, 0)$.

(3) Find $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$.

Solution :

(1) For $(x, y) \neq (0, 0)$, we have

$$f_x(x, y) = \frac{x^4 y + 4x^2 y^3 - y^5}{(x^2 + y^2)^2}, \quad f_y(x, y) = \frac{x^5 - 4x^3 y^2 - xy^4}{(x^2 + y^2)^2}.$$

(2) By definition, we have

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = 0$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-h^5}{h^4} = 0$$

(3) Let

$$g(x, y) = f_x(x, y) = \begin{cases} \frac{x^4 y + 4x^2 y^3 - y^5}{(x^2 + y^2)^2}, & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

$$\Rightarrow g_y(0,0) = f_{xy}(0,0) = \lim_{h \rightarrow 0} \frac{g(0,h) - g(0,0)}{h} = \frac{-h^5}{h} = -1.$$

Let

$$h(x,y) = f_y(x,y) = \begin{cases} \frac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}.$$

$$\Rightarrow h_x(0,0) = f_{yx}(0,0) = \lim_{l \rightarrow 0} \frac{h(l,0) - h(0,0)}{l} = \frac{l^5}{l} = 1.$$

Since $f_{xy}(0,0) \neq f_{yx}(0,0)$ ，則由 Clairaut's Theorem 得知

f_{xy} and f_{yx} 在 $(0,0)$ 附近必不連續。

檢查：If $(x,y) \neq (0,0)$ ，則 $f_{xy}(x,y)$ and $f_{yx}(x,y)$ 皆為分母、分子
為次數 8 的齊次項。

$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} f_{xy}(x,y)$ and $\lim_{(x,y) \rightarrow (0,0)} f_{yx}(x,y)$ 不存在。